

SMART STICK

Submitted By

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24MCD008



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INSTITUTE OF TECHNOLOGY
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SMART STICK

Minor Project

Submitted in partial fulfillment of the requirements

for the degree of

Master of Technology in Computer Science and Engineering (Data Science)

Submitted By

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Certificate

This is to certify that the Minor Project entitled “**SMART STICK**” submitted by **ManjurMahamad Kovadiya (Roll No: 24MCD008)**, towards the partial fulfillment of the requirements for the award of degree of **Master of Technology in Computer Science and Engineering (Data Science)** of Nirma University, Ahmedabad, is the record of work carried out by him under my supervision and guidance. In my opinion, the submitted work has reached a level required for being accepted for examination. The results embodied in this minor project, to the best of my knowledge, haven’t been submitted to any other university or institution for award of any degree or diploma.

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Abstract

The objective of this project is to design a Smart Stick for blind people that assists them to move safely and independently by utilizing a blend of IoT devices and Deep Learning algorithms. The purpose is to offer real-time assistance, object recognition, and voice guidance to make day-to-day movement more secure and well-informed for blind users.

The scope of work includes the implementation of a smart system with **Raspberry Pi (RPI)** to implement various deep learning models, and additionally integration with IoT devices such as **LED light, buzzer** and **ultrasonic sensor**. A mobile application was also implemented with **Google Maps API** to offer voice navigation to enable the user to move from one place to another. The app also helps in offering voice feedback in terms of nearby objects or human beings detected by the system.

The deep learning algorithms used on the Raspberry Pi include several functions such as pedestrian **traffic light** detection, the **YOLO** approach for detection of particular categories such as **pedestrians, cars and traffic lights, crosswalk detection, pot-hole detection, detection of waste bins, detection of streetlights, detection of signboards, and detection of text** for locating names of big places such as schools, shops, and factories as well as brief audio descriptions.

On the IoT side, LED lights show traffic light color: **green LED** glows on a **green pedestrian signal**, **red LED** glows on a **red light** and both flash on a crosswalk. The buzzer beeps upon sensing walls etc. through an ultrasonic sensor. All sensor-integrated and AI-based functionalities run on the RPI board in real time.

A working prototype of the smart stick already exists, demonstrating the integration of smart features to enhance mobility for the blind. The project demonstrates how the integration of cutting-edge AI and IoT technologies can provide practical solutions to real-world accessibility problems.

Abbreviations

RPI	Raspberry Pi
AI	Artificial Intelligence
IoT	Internet of Things
TTS	Text-to-Speech
API	Application Programming Interface
GPS	Global Positioning System
GSM	Global System for Mobile Communications
USB	Universal Serial Bus
LED	Light Emitting Diode
HDMI	High-Definition Multimedia Interface
Wi-Fi	Wireless Fidelity
CSI	Camera Serial Interface
SSD	Solid State Drive
mAP	mean Average Precision
CNN	Convolutional Neural Network
IC	Integrated Circuit
RF	Radio Frequency

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Chapter 1

Introduction

1.1 General

For visually impaired individuals, mobility within shared everyday environments can be challenging and sometimes unsafe. Traditional walking sticks offer only obstacle detection but cannot react with more descriptive feedback regarding the surroundings. With advancements in technology, especially with Internet of Things (IoT) and Artificial Intelligence (AI), there is now immense possibility to provide mobility aids with smart features that can potentially significantly increase blind user independence and safety.

The objective of this project is to build a Smart Stick for usage by visually handicapped people. The stick fuses different algorithms in deep learning and IoT equipment to create a clever and keen system. Raspberry Pi is utilized as the root of the system, which in turn is implemented to carry out input from multivariate sensors and AI models for detecting objects, feeling traffic light, detecting crucial signboards, and providing instantaneous feedback through auditory assistance.

To supplement the user's navigation needs even more, a mobile application has been developed using Google Maps API that gives voice assistance directions and data on nearby places or objects. The stick also includes LED lights, a buzzer, and an ultrasonic sensor for haptic feedback and obstacle detection. All these combined, sync to deliver an even more interactive, informative, and secure trip experience to blind users.

This introduction sets the stage for a comprehensive solution that merges modern technologies to address a critical accessibility issue. The Smart Stick aims not only to guide but also to inform and protect its user, making independent movement more

practical and secure systems.

1.2 Project Aim

The aim of this project is to design and develop a Smart Stick that enhances the mobility and safety of visually impaired individuals by using IoT devices, deep learning models, and voice-assisted technology to detect surroundings and guide users effectively in real-time.

1.3 Objective

- To develop a smart assistive device that helps visually impaired individuals navigate their surroundings safely and independently.
- To integrate Raspberry Pi as the core processing unit to run various deep learning models for object and environment detection.
- To implement models for detecting pedestrian traffic lights, crosswalks, potholes, trash bins, street lights, sign boards, and recognizing text from shop or place names.
- To create a mobile application that uses Google Maps API for voice-guided navigation and location-based assistance.
- To use ultrasonic sensors for obstacle detection and provide real-time feedback using LED indicators and buzzer alerts.
- To provide voice assistance to inform the user about nearby people, objects, and key places, making the system user-friendly and practical.
- To build a fully functional prototype of the Smart Stick to demonstrate the integration and performance of all features.

Chapter 2

Literature Review

The research paper “Development of an IoT-Based Intelligent Guide Stick” proposes a smart stick with different sensors such as ultrasonic sensors, a soil moisture sensor, GPS, and GSM to guide visually impaired users. It offers functionality such as obstacle and water detection, voice warning, and sharing of location. The system is lightweight, low in cost, and offers real-time support, allowing blind individuals to move around independently indoors and outdoors [1].

In “Multi-Functional Blind Stick for Visually Impaired Persons” the authors present a low-cost smart stick with obstacle, stair, and wet surface detection using sensors like ultrasonic, moisture, and infrared sensors. The stick also has location sharing in case of emergencies and a remote control to find a misplaced stick. A mobile app companion allows caregivers to configure emergency contacts [2].

The research paper “WiFi and Bluetooth-based Smart Stick for Guiding Blind People” presents an implementation involving Raspberry Pi, a camera, ultrasonic sensors, and a GPS module. It provides audio navigation through earphones, vibration warning of obstacles, and voice recognition for object detection. The design increases the functionality of blind individuals in terms of understanding and interacting with the surroundings [3].

In “Design and Development of a Low-cost Smart Stick for Visually Impaired Individuals” the proposed model prioritizes obstacle detection and location communication through ultrasonic sensors, GPS, and GSM modules. It includes water detection and stick location based remotely. The device is designed to be cost-efficient, timely, and power-conscious, especially suited for low-income communities [4].

The “Designing Sensor Based Smart Stick for Handicapped Person” paper, presented

at the 2022 ICAC3N Conference, proposes a sophisticated smart stick to enhance mobility for visually impaired individuals, integrating an Arduino Uno microcontroller with multiple sensors, including ultrasonic and infrared for obstacle detection up to 200 cm, a GPS Neo6 module for navigation, and a pulse heart sensor to monitor physical health. The stick provides audio feedback via a voice circuit and vibration alerts, enabling safe navigation indoors and outdoors, while an RF transmitter helps locate the stick if misplaced. Additional features include water and LDR sensors, a SIM800I module for emergency SMS, and a rechargeable battery, though challenges like GPS connection delays and pulse sensor inaccuracies are noted. The system aims to improve independence and safety, with future enhancements like GSM integration and real-time Google Maps coordination to further support visually impaired users [5].

The “Smart Blind Stick” presented at the 2022 ICCUBEA Conference introduces an innovative assistive device for visually impaired individuals, utilizing an Arduino Mega 2560 microcontroller and an ultrasonic sensor (HC-SR04) for obstacle detection within a 22 cm range, with a buzzer and vibrator providing user feedback. The stick operates in two modes: a stationary mode for forward detection and a rotating mode, enabled by a servo motor, for scanning crowded areas up to 60 degrees on either side. While effective, its limited detection range suggests potential enhancements with advanced sensors like infrared or TF Mini, alongside features such as GPS or voice assistance, building on prior research incorporating IoT and autonomous navigation systems to improve safety and independence for visually impaired users [6].

Chapter 3

Tools and Technology

3.1 Hardware Components

3.1.1 Raspberry Pi

The Raspberry Pi 5 figure 3.1 is the latest single-board computer from the Raspberry Pi Foundation, released in October 2023. It's a significant upgrade over the Pi 4, featuring:

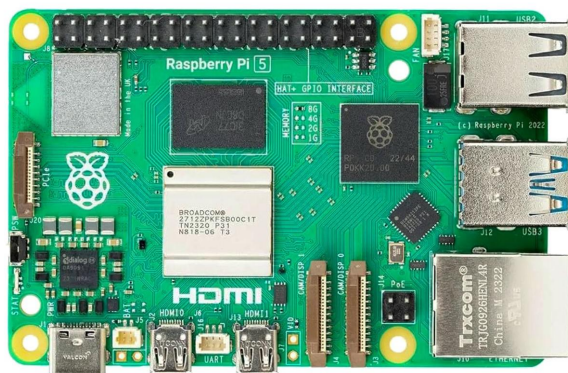


Figure 3.1: Raspberry Pi

- **Processor:** 2.4 GHz quad-core ARM Cortex-A76 CPU (64-bit)
- **GPU:** VideoCore VII graphics processor, supporting dual 4K displays at 60Hz
- **RAM:** 8GB LPDDR4X
- **Storage:** microSD card slot, with PCIe 2.0 support via FPC connector for SSDs
- **Connectivity:** Gigabit Ethernet, dual-band Wi-Fi (802.11ac), Bluetooth 5.0

- **Performance:** Up to 2–3× faster than the Raspberry Pi 4 in many tasks

3.1.2 Pi camera module



Figure 3.2: Pi camera

The Raspberry Pi Camera Module shown in figure 3.2 is a compact, high-quality camera accessory designed for Raspberry Pi boards. It connects via the CSI (Camera Serial Interface) port and supports still photography and HD video recording. This Camera Module v1.3, offers up to 5MP resolution, autofocus, HDR, and improved low-light performance. It's widely used in projects involving computer vision, surveillance, and time-lapse photography. booths, traffic lights, pedestrians, and road-side units (RSUs).

3.1.3 Buzzer



Figure 3.3: Buzzer

A buzzer shown in figure 3.3 is an electronic device that can make a sound. Buzzers are used to alert customers. It is used directly in vehicle transmission and parking machines. It is based on the piezoelectric principle discovered by Jacques and Pierre Curie in 1880.

3.1.4 Ultrasonic Sensor

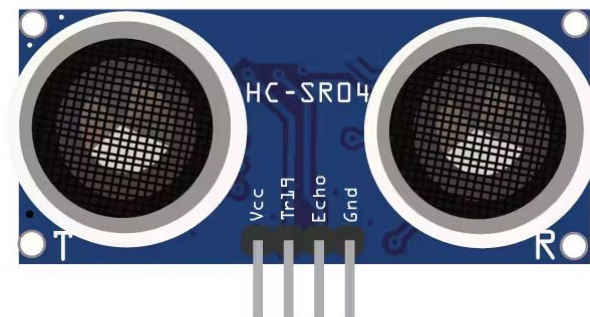


Figure 3.4: Ultrasonic Sensor

An Ultrasonic Sensor shown in figure 3.4 is an electronic device that measures the distance to an object using ultrasonic sound waves. It emits a high-frequency sound pulse and calculates the time it takes for the echo to return, using this to determine the distance. Commonly used in robotics, obstacle detection, and level sensing, it offers accurate, non-contact measurement. The most popular model is the HC-SR04, which provides a range of about 2 cm to 400 cm.

3.1.5 Jumper Wires

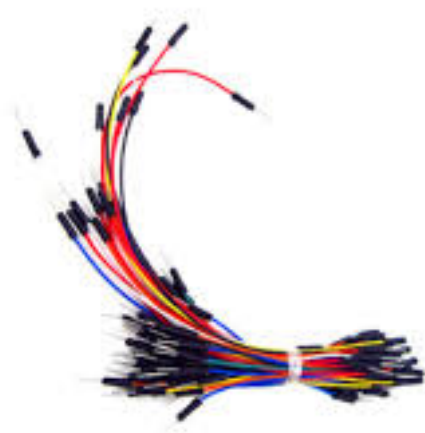


Figure 3.5: Jumper wires

Jumper wires as shown in figure 3.5 are simple wires used to connect different parts of a circuit. They make it easy to join two components without doing any complex wiring or soldering. These wires usually have small pins on the ends that can easily fit into holes on a board or plug into connectors. They are mostly used when building or testing electronic projects on a breadboard.

3.1.6 LED lights



Figure 3.6: LEDs

The LED lights shown in figure 3.6 on the smart stick provide visual feedback for traffic signal detection. When a green signal is detected, the green LED blinks to indicate it's safe to cross; when a red signal is detected, the red LED glows to warn the user to stop. This feature also assists partially sighted individuals and nearby caretakers.

Chapter 4

Methodology

4.1 Workflow

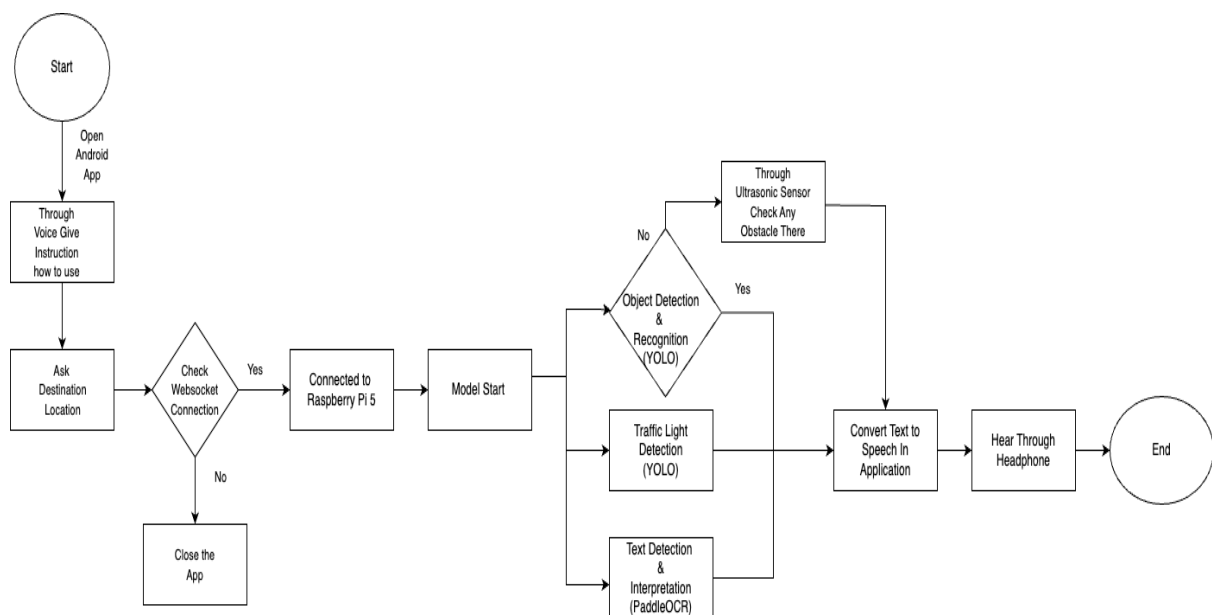


Figure 4.1: Flow Diagram

This figure 4.1 demonstrates the flow of the project. Given below are the explanation steps:

1. Connect Device

The user opens their smartphone and connects it with a headphone or earphone for audio feedback.

2. Launch Smart Stick App

Once the app is opened, it prompts the user to input their destination.

3. **Provide Destination via Voice**

The user speaks the destination address through the connected headphone using voice recognition.

4. **Establish WebSocket Connection**

A WebSocket connection is established between the smart stick and the mobile application to enable continuous, real-time data exchange such as live object detection results and audio responses.

5. **Object and Text Detection in Real-Time**

The smart stick continuously scans the surroundings and detects various objects such as:

- Persons
- Vehicles (car, bus, truck, bicycle)
- Traffic signals (red/green) and Crosswalk
- Signboards and shops
- Text on boards (e.g. shop names and it's brief description)

6. **Signal Detection and LED Feedback**

- If a **green traffic signal** is detected, the **green LED** glows.
- If a **red traffic signal** is detected, the **red LED** glows.

7. **Text-to-Speech Audio Output**

In real time, detected objects and text are converted to speech and played via headphone audio.

Examples:

- “Bus ahead”
- “Medical store on the left”
- “Green signal – you can go”

4.2 Architecture Overview

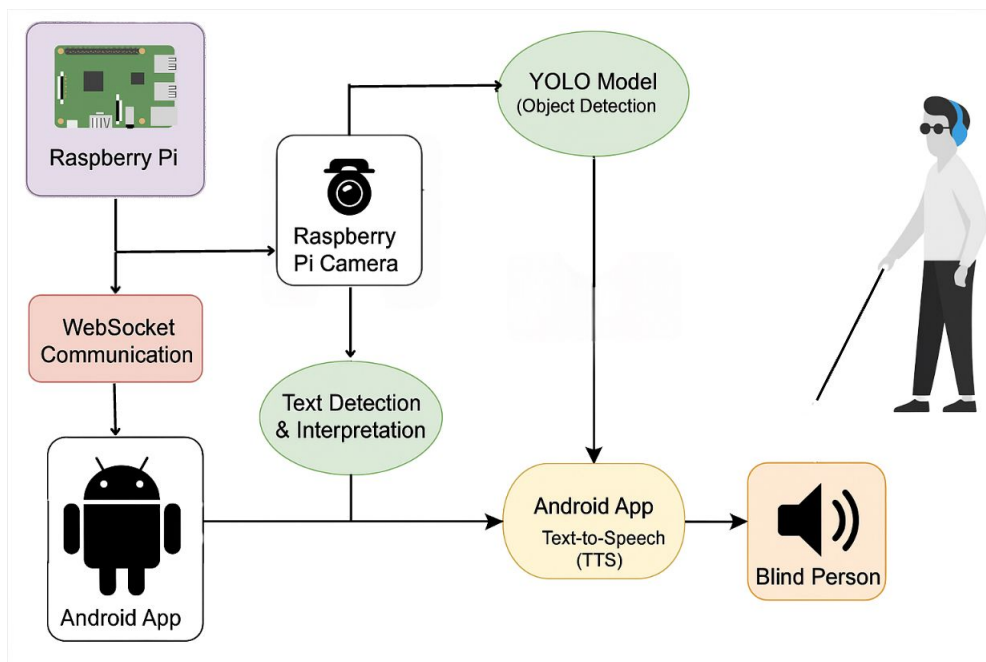


Figure 4.2: Stick Architecture

This architecture 4.2 represents a smart assistive system for visually impaired individuals. At the core of the system is the Raspberry Pi, which is connected to a Raspberry Pi Camera that continuously captures the environment around the user. The camera feed is processed in two main stages: Object Detection and Text Detection & Interpretation.

The YOLO (You Only Look Once) Model is used for real-time object detection. It identifies nearby obstacles like vehicles, pedestrians, and traffic signals. In parallel, the system also performs text detection, recognizing and interpreting text such as signboards or shop names from the captured images.

The processed data (objects and text) is transmitted from the Raspberry Pi to an Android application using WebSocket Communication, which ensures fast, two-way real-time data transfer. The Android app consolidates this information and passes it to a Text-to-Speech (TTS) engine, which converts it into spoken words.

The final output—audio descriptions like "Bus ahead" or "Medical shop to your right"—is delivered to the blind person via headphones, helping them navigate their surroundings independently and safely.

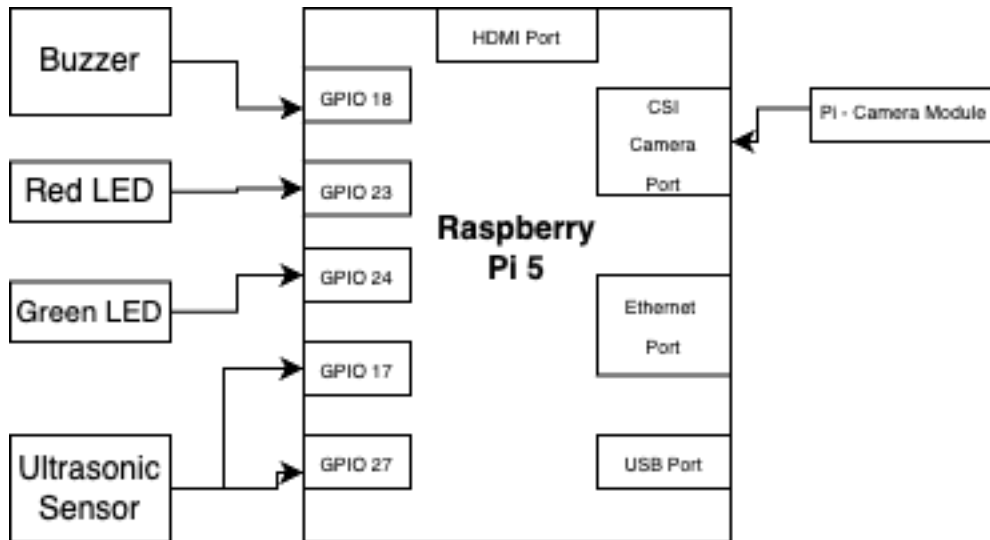


Figure 4.3: Rpi Block Diagram

The image 4.3 shows the Raspberry Pi 5 block diagram, highlighting its connections with various components used in the smart assistive system. The buzzer, Red LED, and Green LED are connected to the Raspberry Pi via GPIO pins 18, 23, and 24 respectively, and are used to give auditory and visual alerts to the user based on environmental inputs. The ultrasonic sensor is connected to GPIO pins 17 and 27 and is responsible for detecting nearby obstacles or changes in distance, ensuring safer navigation. The Pi Camera Module is connected through the CSI Camera Port, which enables the Raspberry Pi to capture images and live video for object and text detection using computer vision models like YOLO. The diagram also shows standard ports like HDMI, Ethernet, and USB, which can be used for display, internet connectivity, and peripheral device connections. This overall hardware architecture forms the backbone of the smart stick system designed to aid visually impaired individuals in real-time.

Chapter 5

Implementation

5.1 Mobile Application

5.1.1 Android Application and Raspberry Pi Integration

1. Overview

The project integrates a custom Android application with a Raspberry Pi to support visually impaired individuals in navigation and environmental awareness. The system uses a combination of WebSocket, Google Text-to-Speech, and Google Maps API to create a smart, interactive experience between the user and the system components.

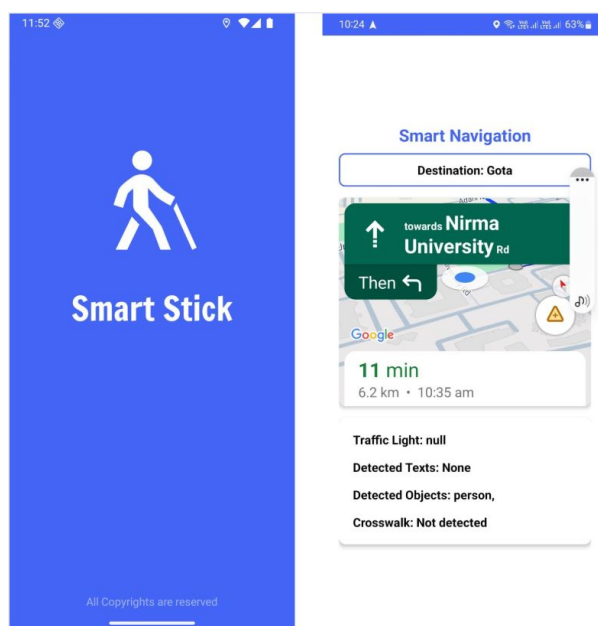


Figure 5.1: Mobile Application

2. Android Application Features

The Android application serves as the user interface and control hub for the smart assistance system. Key functionalities include:

- **WebSocket Communication:**

Establishes a real-time, bidirectional communication channel with the Raspberry Pi. The app sends commands and receives processed data such as obstacle alerts or text recognized from surroundings.

- **Google Text-to-Speech (TTS)**

Converts important messages (e.g., obstacle detection, navigation directions) into audio feedback, ensuring real-time, clear communication with the user.

- **Google Maps API Integration:**

Provides navigation assistance using GPS and real-time route tracking. The application guides the user to desired destinations with voice navigation and safe routing.

- **Voice Command Support:**

The user can interact with the application through voice commands to control functions such as setting destinations, requesting surrounding text, or checking obstacles.

3. **WebSocket Connection to Raspberry Pi**

The Raspberry Pi hosts a WebSocket server that listens for connections from the Android device. The real-time communication allows for:

- Sending image data or sensor information to the app.
- Receiving control commands such as “Start Navigation,” “Read Surroundings,” or “Detect Obstacles.”
- Fast and reliable data exchange suitable for assistive devices requiring low-latency interaction.

4. **Google Text-to-Speech API**

The Android app leverages Google Text-to-Speech to provide real-time audio responses to the user. This includes:

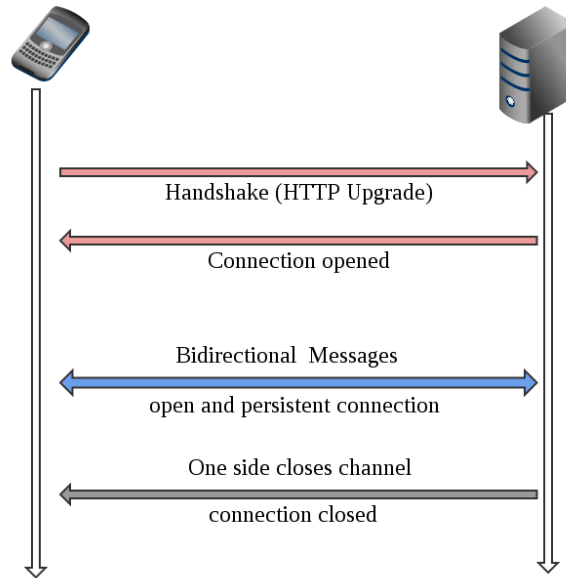


Figure 5.2: WebSocket Architecture

- Announcing obstacle alerts
- Reading detected text from surroundings
- Giving turn-by-turn directions

TTS ensures the visually impaired user is always informed through natural-sounding voice feedback.

5. Google Maps API for Navigation

The app integrates Google Maps API to enable GPS-based path finding and navigation. Features include:

- Setting destination via voice or touch
- Route calculation with pedestrian-safe paths
- Real-time turn-by-turn audio directions
- Location tracking using the Raspberry Pi GPS or the phone's location

This provides the user with reliable and context-aware navigation capabilities.

6. Benefits for Visually Impaired Users

- Hands-free control through voice interaction.

- Real-time environmental feedback using TTS.
- Safe and accurate GPS navigation.

5.2 Text Processing

5.2.1 Text Detection and Interpretation

1. Overview

One of the key features of this project is the ability to detect and interpret text from the environment in real time, enabling blind or visually impaired users to understand their surroundings. For this purpose, we used PaddleOCR, an advanced deep learning-based Optical Character Recognition (OCR) library, in combination with the Google Places API for semantic interpretation of detected text (e.g., identifying nearby shops, landmarks, or signs).

2. Text Detection Using PaddleOCR

(a) PaddleOCR Overview

PaddleOCR is an open-source, high-accuracy OCR system developed by Baidu. It supports more than 80 languages and provides a complete pipeline for text detection and recognition. In this project, we used the English OCR lightweight model for efficient performance on edge devices like the Raspberry Pi.

(b) Dataset Used

PaddleOCR is trained on a variety of publicly available datasets, including:

- ICDAR 2015: Contains street view images with various text orientations, commonly used for scene text detection and recognition.
- SynthText: A large-scale synthetic dataset used to pre-train models on diverse text styles and environments.
- COCO-Text: Annotated natural images containing machine-printed and handwritten text.

These datasets help PaddleOCR recognize diverse text styles under real-world conditions. PaddleOCR pre-trained models were directly utilized in this project,

allowing us to avoid training from scratch and focus on integrating the pipeline for real-time usage.

(c) Text Detection Pipeline

- i. Image Capture: The Raspberry Pi camera captures frames upon user request or automatically at intervals.
- ii. Preprocessing: Images are resized and enhanced for optimal detection.
- iii. Detection: PaddleOCR locates bounding boxes around text regions.
- iv. Recognition: The model extracts the readable text from the detected regions.
- v. Result Transfer: The recognized text is sent to the Android app via Web-Socket for interpretation.

3. Text Interpretation Using Google Places API

The detected text often includes names of nearby places, shop boards, or road signs. To provide contextual meaning to the user, the system interprets this data using the Google Places API.

(a) How It Works

- Send Detected Text and Location: The recognized text and GPS coordinates are sent to Google Places API.
- Search Nearby Places: The API returns matching businesses or landmarks near the current location.
- Determine Relevance: If the text corresponds to a known place, additional information is retrieved (e.g., place type, opening hours).
- Voice Feedback to User: Google Text-to-Speech reads out the relevant interpretation.

(b) Example Scenario

- Detected Text: "Apollo Pharmacy"
- Current Location: [18.5204, 73.8567]
- Google Places API Response:
 - Name: Apollo Pharmacy

- Type: Pharmacy
- Status: Open Now
- Rating: 4.3

Spoken Output: “You are near Apollo Pharmacy. It is currently open.”

4. Benefits for Visually Impaired Users

- Contextual Awareness: Helps users recognize nearby shops, public signs, and landmarks.
- Real-time Interpretation: Detected text is not only read but interpreted for meaningful interaction.
- Local Relevance: Place matching ensures that only relevant nearby information is delivered.
- Safety and Independence: Reduces dependence on others by making the user aware of the environment.

5. Summary

By combining PaddleOCR for accurate text detection and Google Places API for meaningful interpretation, the system ensures that visually impaired users can understand and navigate their surroundings efficiently.

5.3 Prototype



Figure 5.3: Smart Stick Prototype

The figure 5.3 shows a smart stick prototype designed for visually impaired users. It features an ultrasonic sensor for obstacle detection, LED indicators, and a microcontroller (like a Raspberry Pi) enclosed in a transparent box mounted on a stick. The setup is powered through connected cables and is shown in a lab environment.

Chapter 6

Results

6.1 Model 1: Streetlight, Trashbin and Pothole Detection

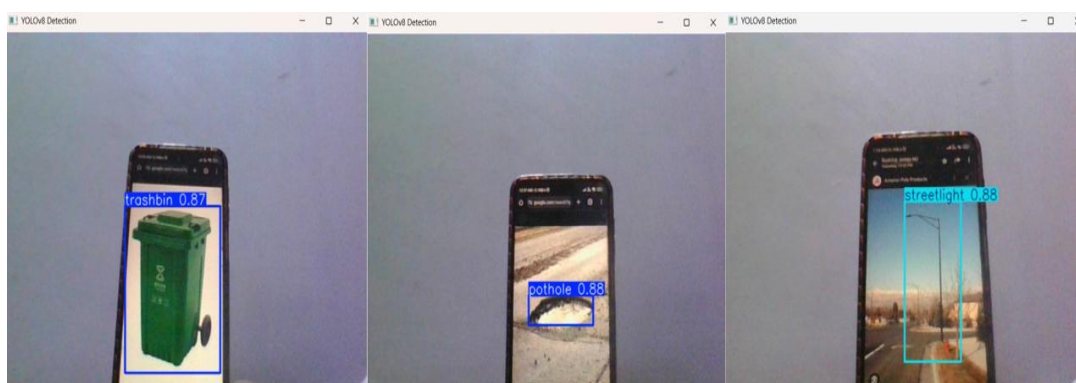


Figure 6.1: Streetlight, Trashbin and Pothole Detection

The figure 6.1 shows YOLOv8 detecting three custom objects—trash bin (0.87), pothole (0.88), and streetlight (0.88)—from photos displayed on a phone screen. Each object is correctly labeled with a bounding box and confidence score, indicating successful custom class detection.

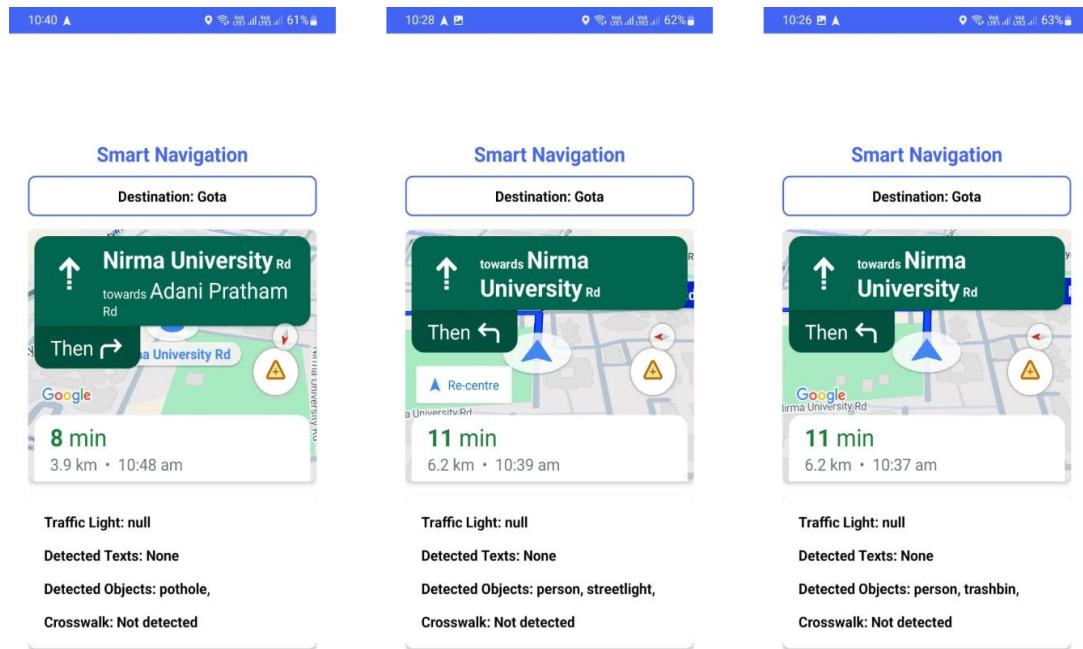


Figure 6.2: Street Light, Trashbin, Pothole results in Application

The figure 6.2 shows a Smart Navigation app detecting objects during navigation to Gota. It identifies pothole, person, streetlight, and trashbin across different screens, while crosswalk, text, and traffic lights are not detected.

6.2 Model 2: Pedestrian Traffic Light and Crosswalk Detection

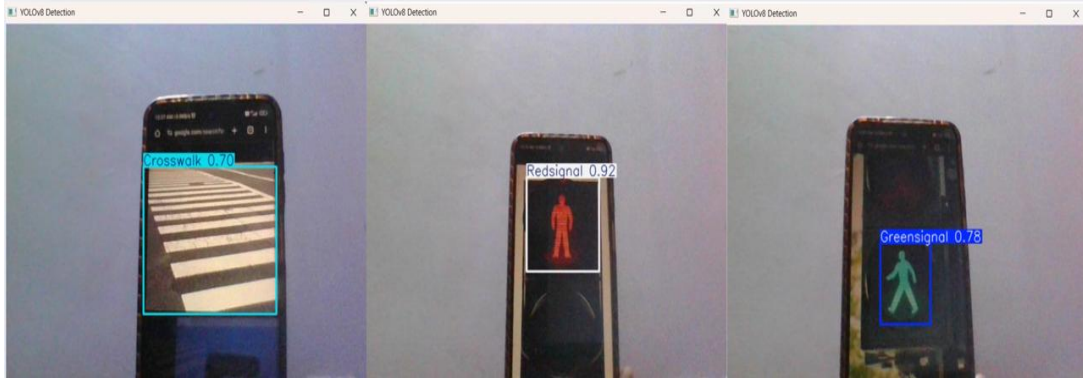


Figure 6.3: Pedestrian Traffic Light and Crosswalk Detection

The figure 6.3 shows YOLOv8 detecting three custom objects—crosswalk (0.70), red signal (0.92), and green signal (0.78)—from images displayed on a phone screen. Each object is accurately labeled with a bounding box and confidence score.

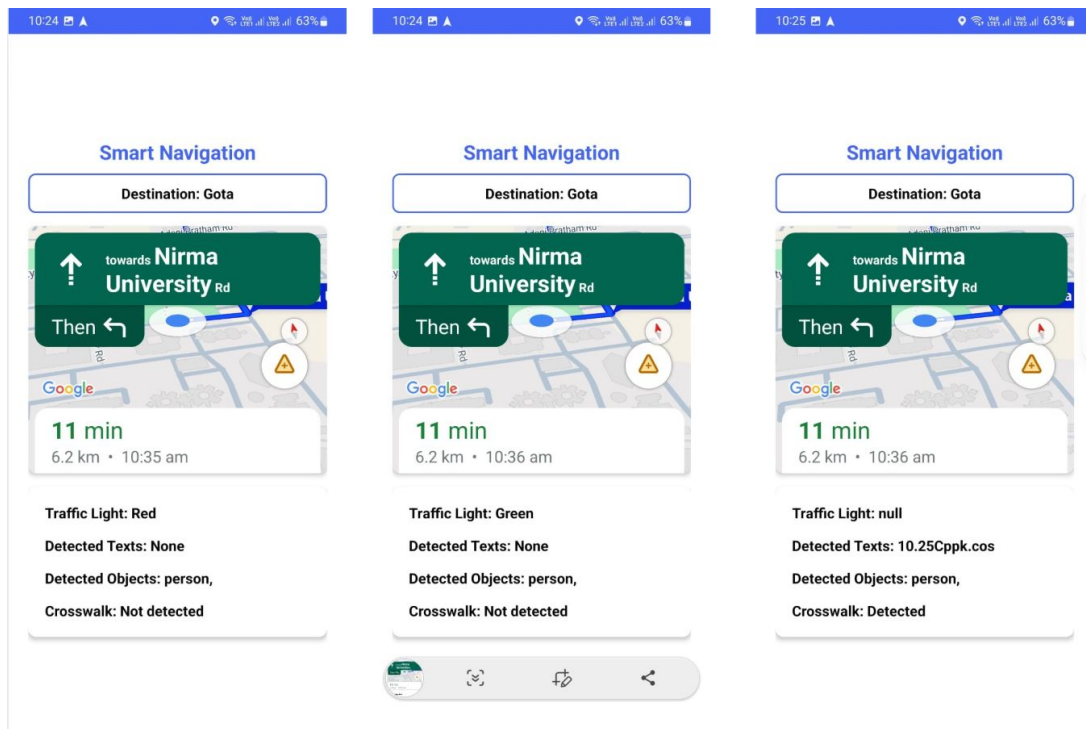


Figure 6.4: Pedestrian and Crosswalk results in Application

The figure 6.4 shows three screens. The first screen shows a red traffic light, while the second shows a green one. The third screen does not detect any traffic light but

successfully identifies a crosswalk and some text. This demonstrates the app's ability to recognize various elements relevant to pedestrian safety.

6.3 Model 3:Text Detection

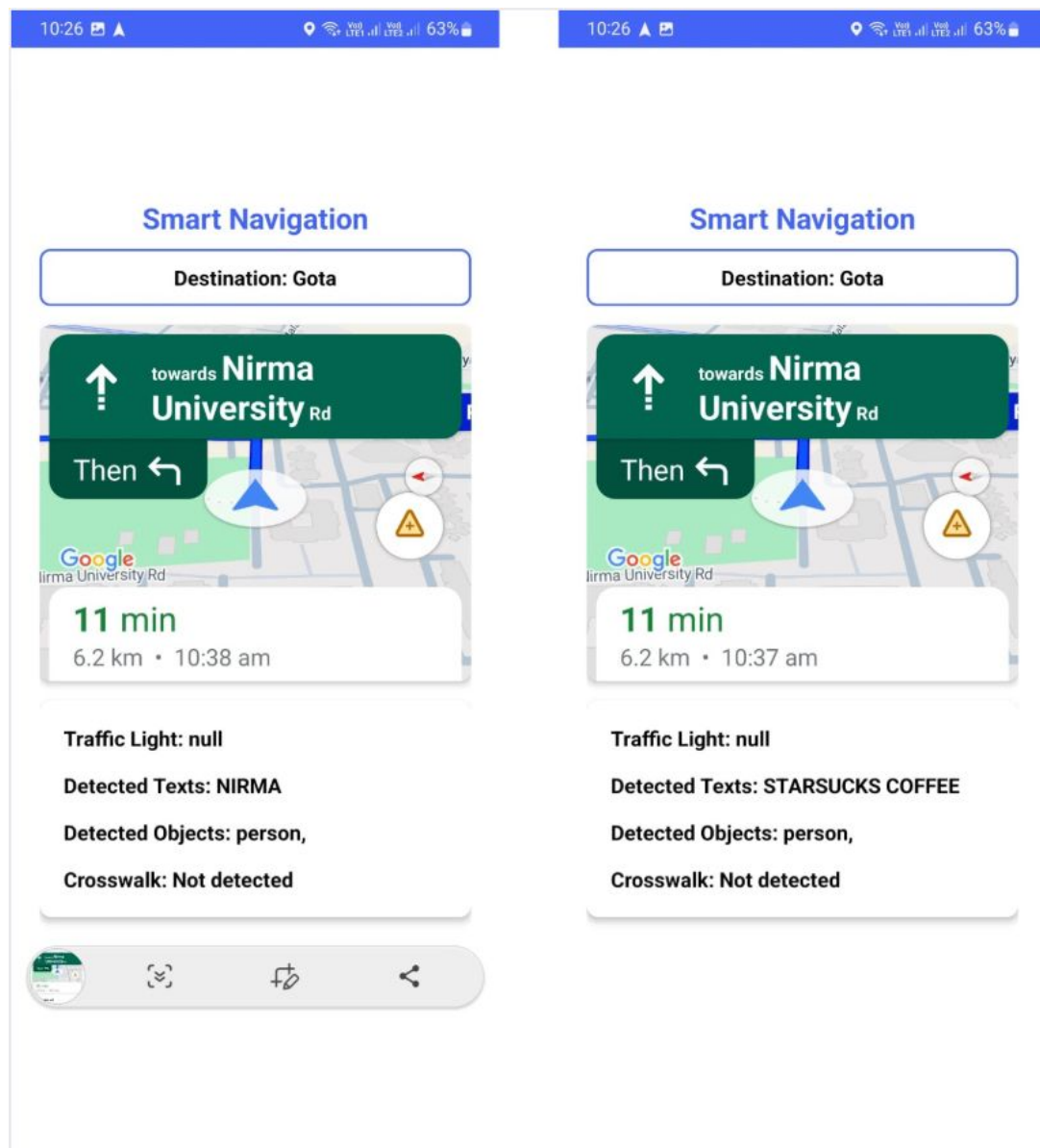


Figure 6.5: Store Name and Places Name results in Application

The figure 6.5 show the smart navigation app interface showing a 6.2 km, 11-minute route to "Gota" with directions toward Nirma University Rd. Detects nearby objects (person) and texts ("NIRMA" or "STARBUCKS COFFEE"), with no traffic lights or crosswalks noted.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

The Smart Stick project was initiated with the goal of enhancing visually impaired individuals' mobility and autonomy by integrating the latest technologies such as IoT, deep learning, and mobile support. By a systematic and research-based approach, we were successful in designing, training, and deploying an operating prototype that is capable of sensing and responding intelligently to actual-world obstacles and navigation needs.

The project began with the collection and preparation of the datasets required to train deep learning models for pedestrian traffic light detection, potholes, trash cans, signboards, crosswalks, and text capture and recognition from a variety of landmarks. Through the application of YOLO and transfer learning techniques, we could develop models capable of real-time object and element detection with reasonable accuracy.

Alongside the AI models, we developed a mobile application using Google Maps API for navigation directions and voice announcements to inform the user of nearby objects or persons. The voice announcement mechanism played a critical role in enabling the user to receive valuable, real-time information without needing a display.

The final hardware prototype was constructed by placing all the most crucial elements including Raspberry Pi, ultrasonic sensors, LED indicators, buzzers, and power modules inside a stick structure to produce a compact and user-friendly supportive device.

Overall, the Smart Stick combines the power of embedded systems and artificial intelligence to solve practical problems faced by the visually impaired. The proof-of-concept serves as an instance of a scalable and viable solution that can be further expanded upon

and possibly mass-produced to make navigation safer and easier for those in greatest need.

7.2 Future Work

- **Distance Measurement:** Add functionality to measure and announce the distance of detected objects, helping users estimate how far an obstacle or person is.
- **Activity Detection:** Implement deep learning models to detect and alert users about abusive or suspicious activities in the surroundings.
- **Advanced Water Detection:** Introduce water-level sensing and the ability to detect spiral water movement, identifying flowing or potentially dangerous water bodies.
- **Haptic Feedback:** Integrate vibration patterns (haptic feedback) to provide silent, intuitive alerts based on different types of detections.
- **Find My Stick:** Add a feature using Bluetooth and GPS to help users locate their stick if misplaced.
- **Step/Staircase Detection:** Enable detection of steps and staircases to avoid trips or falls.
- **Model Accuracy Improvement:** Enhance AI model performance using better datasets and advanced architectures such as YOLOv8.
- **Real-Time Face Recognition:** Identify known individuals or alert when unknown persons are nearby.
- **Weather Update Integration:** Provide current weather alerts through the voice assistant to help users prepare for conditions.

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